

2.4 Rank-Nullity Theorem

2.4.1 Nullity and Rank

Definition 2.4.1 — Nullity

Let $f : V \rightarrow W$ be a linear transformation. The **nullity** of f is defined by

$$\text{nullity}(f) := \dim(\ker(f)).$$

Definition 2.4.2 — Rank

Let $f : V \rightarrow W$ be a linear transformation. The **rank** of f is defined by

$$\text{rank}(f) := \dim(\text{im}(f)).$$

Example.

Recall that for $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x),$$

we have $\ker(D) = \mathbb{R}$ and $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$. Compute the nullity and image of D .

Solution. $\text{nullity}(D) := \dim_{\mathbb{R}}(\ker(D)) = \dim_{\mathbb{R}}(\mathbb{R}) = 1.$

$\text{rank}(D) := \dim_{\mathbb{R}}(\text{im}(D)) = \dim_{\mathbb{R}}(\mathbb{R}[x]_{\leq 2}) = 3.$

Remark. For other examples of nullity and rank, review the examples in the last section, which we have it covered.

2.4.2 The Theorem

Theorem 2.4.1 — Rank-Nullity Theorem

Let V be a finite dimensional vector space and $f : V \rightarrow W$ be a linear transformation. Then we have

$$\text{nullity}(f) + \text{rank}(f) = \dim(V).$$

Proof. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$, having $\text{nullity}(f) = \dim(\ker f) = n$. Since $\ker(f) \subseteq V$, by Basis Extension Theorem, we can extend S to $S' = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\}$, a basis of V , having $\dim(V) = n + k$.

Claim. $S'' = \{f(\mathbf{s}_1), f(\mathbf{s}_2), \dots, f(\mathbf{s}_k)\}$ is a basis of $\text{im}(f)$.